

## Technical Datasheet of RS485 over BLE5

### Product Overview: -

**WIN-GW-BLE5-RS485-T** is an industrial-grade bi-directional **RS-485 to BLE** Transparent Bridge, designed to bridge legacy serial devices with modern IoT and BLE networks. Featuring a **built-in Web UI**, it allows easy configuration of **RS-485 communication parameters**. This compact and rugged device ensures **reliable data transmission** from serial devices to BLE platforms using the Transparent BLE protocol.



## Key Features:

1. **RS485 Serial Interface**  
Supports long-distance serial communication (up to 1200 meters) with industrial devices.
2. **BLE Protocol Support**  
Publishes Transparent BLE↔RS485 gateway supporting custom commands and raw data exchange.
3. **BLE Wireless Communication Range (extendable)**  
Provide reliable BLE connectivity up to 20–30 meters in **open line-of-sight (LOS)** environments and 10–15 meters in **non-line-of-sight** (indoor/obstructed) conditions, depending on installation and interference levels.
4. **Web-Based Configuration**  
Intuitive Web UI for setting RS485 parameters as Baud/SID/Parity.
5. **Real-Time Data Publishing**  
Ensures low-latency communication to client dashboards.
6. **Compact & Industrial-Grade Design**  
Suitable for deployment in harsh environments.

## Advantages:

1. Transparent BLE↔RS485 Communication
2. Provides stable and high-speed data transfer for industrial environments.
3. Low-Latency Data Publishing
4. Real-time data delivery ensures accurate and timely monitoring.
5. Allows multiple BLE devices to connect simultaneously for monitoring, control, and diagnostics.
6. Compatible with custom mobile apps, BLE gateways, and embedded BLE hosts, for raw data transport.
7. Supports BLE 5.0 connectivity.
8. Offers a cost-effective way to modernize wired serial systems with wireless BLE connectivity.

### Specifications:

1. **Dimensions :** 70 mm L x 110 mm B x 50 mm H
2. **Power :** 12 DC, typical power consumption – 5W
3. **Operating Temperature :** 0 – 60° C (32 ~ 140°F)
4. **Storage Temperature :** -20 - 70° C (-4 ~ 158°F)
5. **Storage Humidity :** 5 ~ 95 % RH, non – Condensing
6. **Certifications :**



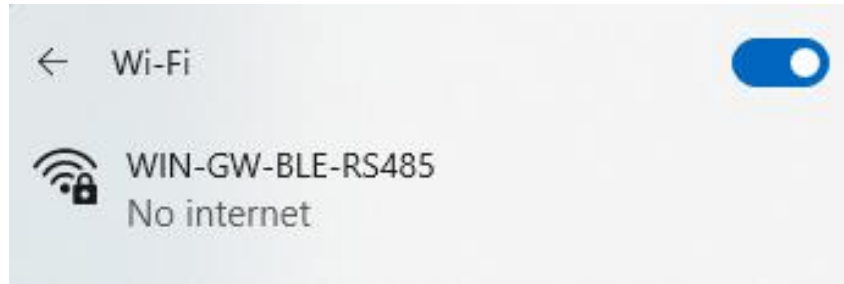
### Additional Features: -

- Communication Port isolated Input power reverse polarity safety
- ESD Safety IEC 61000-4-2,  $\pm 30$ KV contact,  $\pm 30$ KV air
- EFT IEC 61000-4-4, 50A (5/50ms) 750V isolation.
- CRC Error check.

### WiFi Configuration Settings: -

- Connect a 24V AC/DC power supply to the module.
- Connect one or more RS485 slave devices/RS485 terminal to the module's A & B terminals.
- Then Access the module WIFI and enter the IP: 192.168.4.1 to configure the RS485 setting.

(WIFI: **WIN-GW-BLE-RS485**, PASS: **12345678**).



### WittelB BLE-RS485 Gateway

Configure your RS485 communication settings

**Baud Rate**

9600

**Data Bits**

8

**Parity**

Even

Save Config

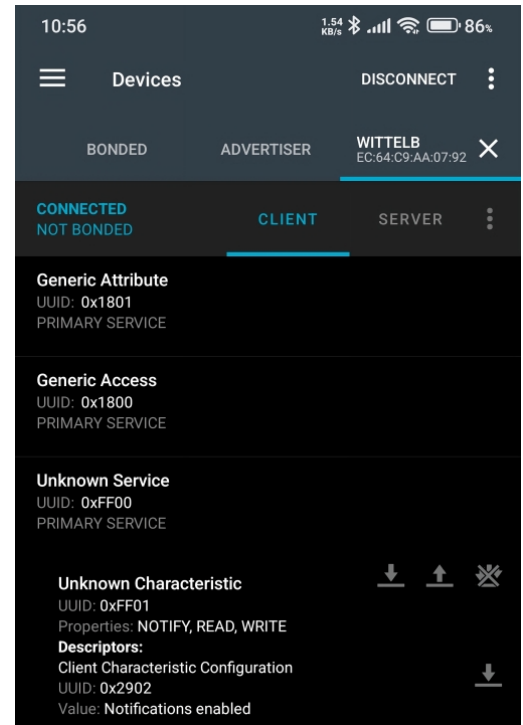
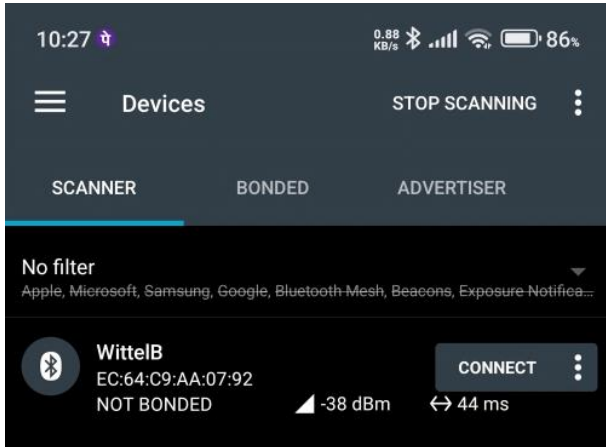
Reset

WiFi: WIN-GW-BLE-RS485  
IP: 192.168.4.1

- Enter the **Baud Rate**, **Data Bits**, **Parity**, according to your communication settings.
- Here you can configure the serial communication parameters for the device you are setting up.

## 1. BLE Client to RS485 Terminal:-

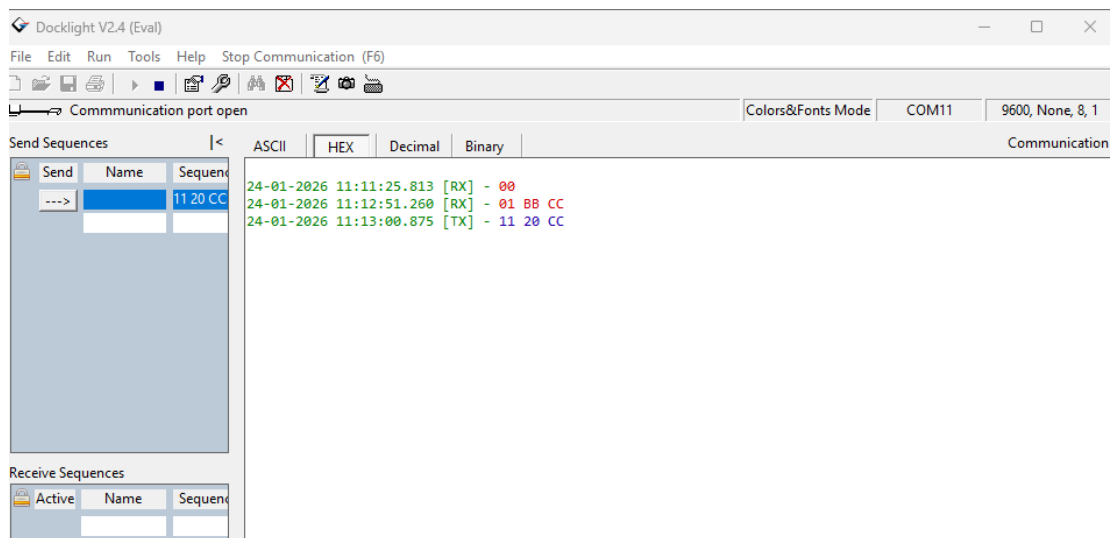
- Open BLE Client Apps (ex: NRF Connect) or customized BLE Applications.
- Connect with BLE ID WittelB.



- After Connection enable the notification to read the packets.
- Also, you can write the commands for RS485 in hex.
- It supports 12 bytes and more.

## 2. RS485 Terminal to BLE Client:-

- Open RS485 Terminal (ex: Docklight) and connect with port and default RS485 address as baud: 9600, Data bit: 8, Parity: None.



- Then send hex commands from the RS485 Terminal, which will appear in BLE Client App as transparent.
- And also receive hex commands which is sent from the BLE Clients as transparent.
- It supports 12 bytes and more.

### **3. BLE Client to RS485 Slaves:-**

- Open BLE Client Apps (ex: NRF Connect) or customized BLE Applications.
- Connect with BLE ID WittelB.
- Refer Image1&2 for BLE Connections.
- After Connection enable the notification to read the data packets of the slaves.
- Then write the command for RS485 slave in hex to read data packet.

#### **Contact us: -**

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